

Pharmaceutical Analytical Chemistry II

PCC/PCD 105

Week 6 lecture

Cerium (IV), Ce^{4+}

- **Properties**

- Strong Oxidizing agent, Cerium (IV) ammonium nitrate, $\text{Ce}(\text{NH}_4)_2(\text{NO}_3)_6$
- Primary standard
- doesn't oxidize chloride... no need for ZR reagent

- **Stability of standard solution**

- Salt is dissolved in sulfuric acid, at least 0.1 M, to prevent precipitation of **basic salts**.

(Ceric salt hydrolyzes to form **ceric hydroxide** if the solution is not acidic)

- Solution in sulfuric acid is indefinitely stable.
- Solution in nitric acid undergoes **photo-chemical decomposition**

- **End point detection**

Ferriin indicator From red to blue

Reduction half-reactions



Yellow

Change in oxidation state of Cerium: +4 *to* +3

Application

By direct titrations

Determination of reducing agents:

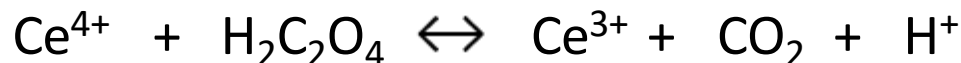
- Ferrous salts



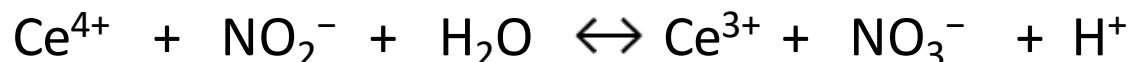
- Hydrogen peroxide



- Oxalic acid



- Nitrous acid



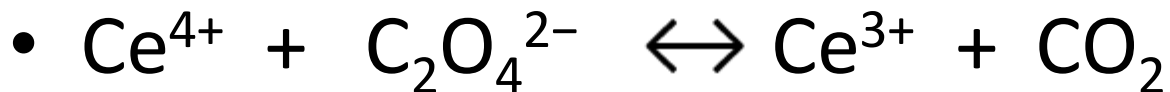
By back titration

- **Determination of Glycerol**



$$\text{Eq. Wt} = \text{M.wt} / 8$$

back titrate xss unused ceric using st. Oxalate, ferroin ind.



Color change: Ferriin (blue) to Ferroin (red)

Potassium dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$

- **Properties**

- Slightly weaker oxidizing agent than ceric and MnO_4^-
- It does not oxidize Cl^- ∴ No need for ZR reagent
- Its reactions are slow (Back titration in most cases is preferred)
- Can not oxidize oxalic or ferrocyanide.
- Primary standard, its solution does not need to be standardized.
- Titrations are carried out in acidic solution (WHY)

In *basic* solution, $\text{Cr}_2\text{O}_7^{2-}$ is converted to yellow chromate ion (CrO_4^{2-}), whose oxidizing power is nil.

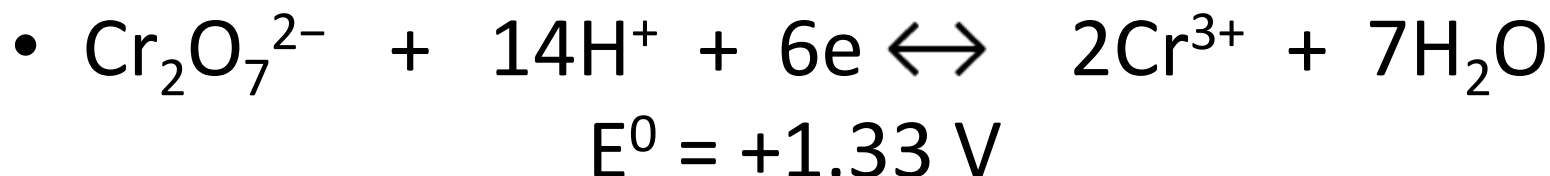
- **End point detection**

Not self indicator

Ferroun indicator From red to blue

Diphenylamine sulphonate from colorless to violet

Reduction half-reactions



- Orange**

green

$$2 \text{ Cr} = +6 \times 2 = +12$$

$$2 \times +3 = +6$$

- Change in oxidation state of 2Cr (1mole): +12
to +6
- Eq. wt = Mwt / 6 for 1 mole (2Cr)

Application

By direct titrations

Determination of reducing agents:

- Ferrous salts



- A mixture of sulfuric / phosphoric acid is added in this titration.

- H₃PO₄

It complexes Fe³⁺ to (Fe(HPO₄)⁺), accordingly it will:

Mask its yellow color → sharp end point

↓ the E of Fe³⁺/Fe²⁺ system → ↓ reduction liability of Fe³⁺/Fe²⁺ system

↑ oxidation liability (reducing power) of Fe³⁺/Fe²⁺ system

↑ feasibility of titration of Fe²⁺ against Cr₂O₇²⁻

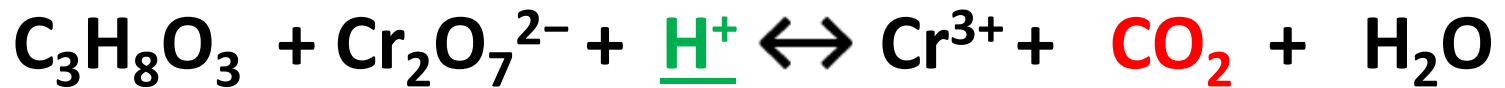
- H₂SO₄: for acidification

- Ind: diphenylamine sulfonic acid

Color is changed from colorless to violet

By back titration

- **Determination of Glycerol**



$$3\text{C} = -2$$

$$1\text{C}=+4 \text{ (i.e. } 3\text{C}=+12)$$

- -2 to +12 ----- loss of 14 e
- Eq wt glycerol = = Mw / 14

In a tabular form compare between:

	Permanganate	Ceric	Dichromate
Strength	$E^0 = +1.51 \text{ V}$	$E^0 1.44 \text{ V}$ (1 M H_2SO_4)	$E^0 = +1.33 \text{ V}$
Standard solution	secondary	primary	Primary
Reaction with chloride	+ve ZR reagent	-ve	-ve Sulphuric/ phosphoric
Detection of e.p.	self	ferroin	Ferroin Diphenyl amine
Applications: 1-direct	Determination of reducing agents, in acidic medium		
2- Back application	Determination of glycerol, alkaline medium	Determination of glycerol, acidic medium	

- In a tabular form, compare between determination of glycerol using different oxidizing agents.

	Permanganate	Ceric	Dichromate
Medium	Alkaline	Acidic	acidic
Eq. wt of glyc.	Mwt/14	Mwt/8	Mwt/14
End product of glycerol (sample)	Carbonate	Formic acid	Carbon dioxide
End product of standard	Manganate (Ba^{2+})	cerous	chromium
Back titration reaction, indicator			
precautions	Ba^{2+} is added	-----	heat